

Pickleball Shot & Rally Analysis

Benjamin Helton | [Tableau Dashboard](#) | [GitHub Repository](#)

An end-to-end analytics pipeline using data from over 300,000 pickleball shots from matches at all skill levels, combining SQL, predictive modeling, and AI-generated insights. Built as a portfolio project tied to my former position as the Treasurer of the University of Tennessee Pickleball Club, applying the same exploratory-to-predictive workflow used in financial analysis (data validation, trend analysis, predictive modeling, stakeholder-ready reporting) to a new domain. Built using Claude as a coding and analysis tool; all modeling decisions, validation, and data interpretation below are my own.

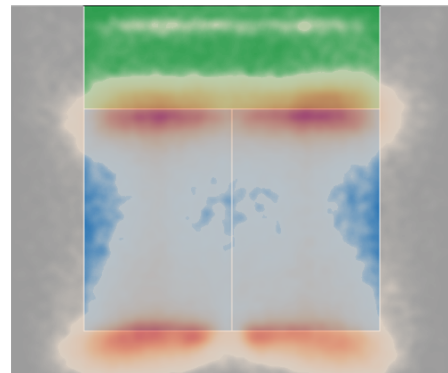
Data and Tools

Source: [Kaggle's PKLMarts](#) dataset, which includes seven relational tables and data on 304,650 shots.

Tools: Claude, SQL/SQLite, Python (pandas, scikit-learn), Tableau Public, and Anthropic API

What I Built

- Data Pipeline: loaded and joined a 7-table relational dataset
- Exploratory Analysis: key rally and outcome trends by skill level
- Spatial Analysis: shot location heatmap (pictured) built using Python and Tableau, validated against real court dimensions
 - Identified and corrected key data interpretation issue: shot coordinates are recorded from hitting player's perspective as opposed to fixed court position, which initially made data look asymmetric
- Predictive Modeling: Built a logistic regression model predicting whether a rally ends in a winner or an error using rally length, shot type, and location as features (80/20 train/test split). Applied class weighting after the initial model favored the majority class.
- AI Integration: Used the Anthropic API to generate natural language scouting reports from shot statistics. Revised the prompt after the first version presented all skill levels with equal confidence regardless of sample size; the final version explicitly flags any group under 30 shots as statistically unreliable.



Key Findings

- Average rally length increases steadily with skill level, from 4.2 shots at the 2.5 level to 10.1 shots at the Pro level.
- A naive baseline (always predicting the majority outcome) achieves 66% accuracy. My initial model reached 71% accuracy but caught only 41% of actual winning shots. Applying class weighting traded overall accuracy for recall, catching 78% of winning shots at a lower precision, a tradeoff worth understanding before treating headline accuracy as the right metric.
- Defensive Reset shots end rallies in a winner 48% of the time, nearly matching aggressive Speed Up shots (54%), suggesting that neutralizing an opponent's attack can be as effective as attacking directly
- A concentrated cluster of shots struck between 1 to 2 feet from the net (as seen on heatmap), dominated by Hand Battle and Dink shots, which reflects the close-quarter net exchanges that decide many competitive points

Reflection

This project allowed me to apply SQL, Python, predictive modeling, and AI prompt engineering to a real dataset. Several of the most useful findings came from questioning results that did not initially align with my previous personal knowledge of pickleball, such as the coordinate system and the effectiveness of reset shots, rather than accepting the data at face value. That process of validating findings and communicating limitations clearly reflects how I approach analytical work.